MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT, GENERAL LIGHTING

50 WATTS, 100 VOLTS, WATERTIGHT, SYMBOLS 92.2 AND 93.2

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SH).

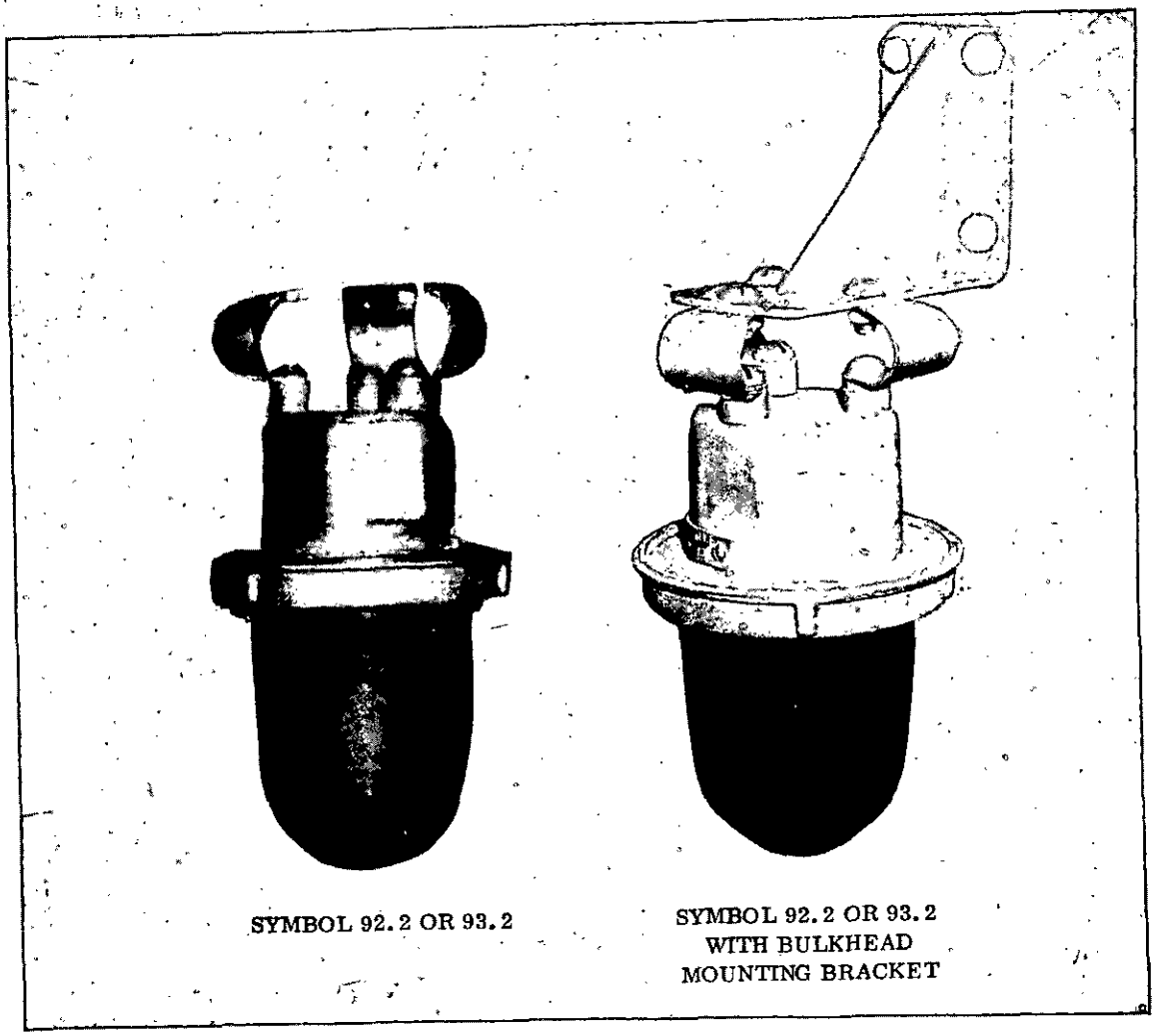


FIGURE 1. Fixtures, Incandescent, Symbols 92.2 and 93.2.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

FSC 6210

Source: https://assist.dla.mil -- Downloaded: 202640-18T08:45Z
Check the source to verify that this is the current version before use.

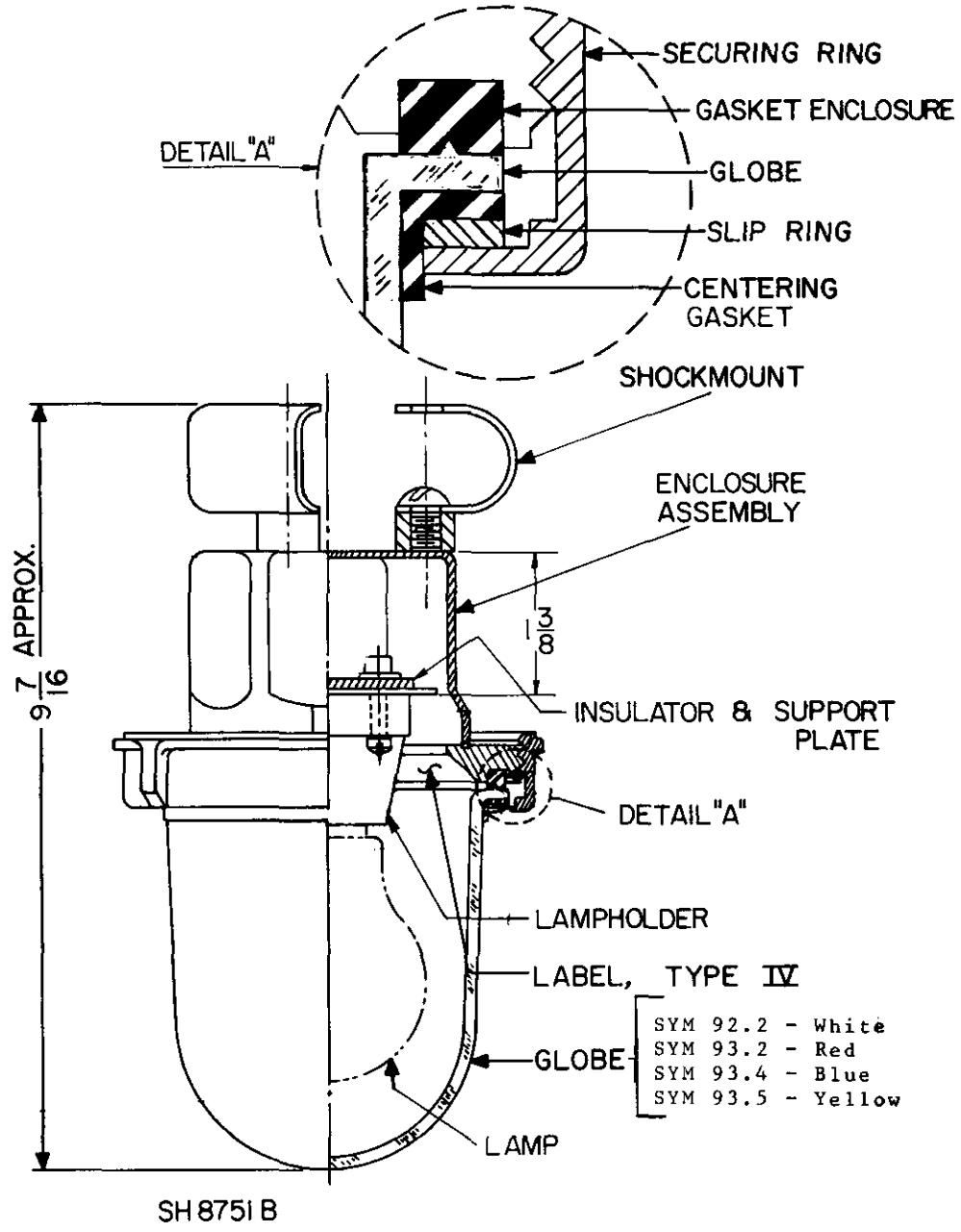


FIGURE 2. Fixture assembly.

Source: https://assist.dla.mil -- Downloaded: 202610-18T08:45Z
Check the source to verify that this is the current version before use.

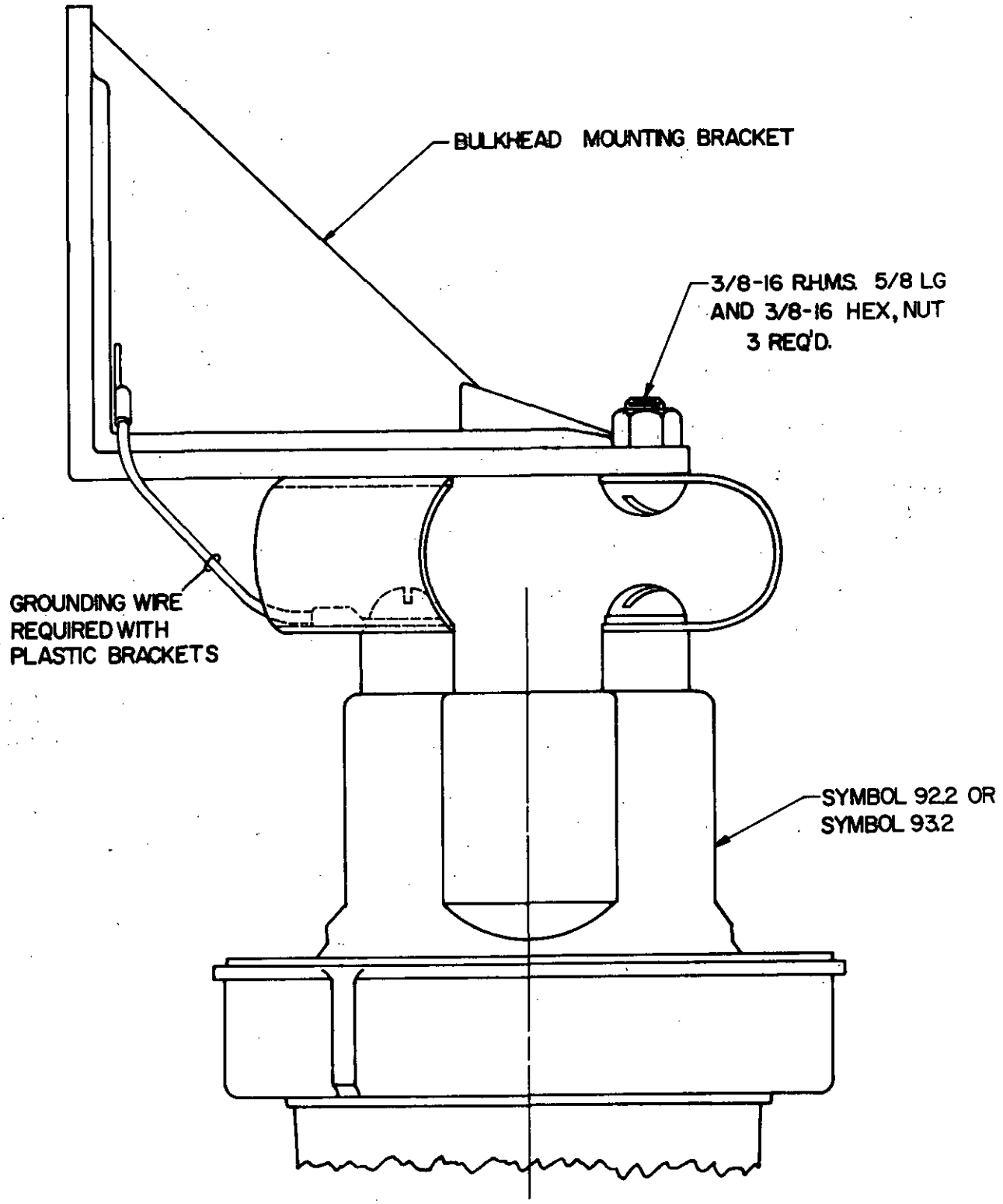


FIGURE 3. Fixture assembly for bulkhead mounting.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:45Z
Check the source to verify that this is the current version before use.

DIMENSION	BRONZE OR BRASS RING	PLASTIC RING
A	13/16	1.190
B	1/16	0.190
C	7/16	0.800

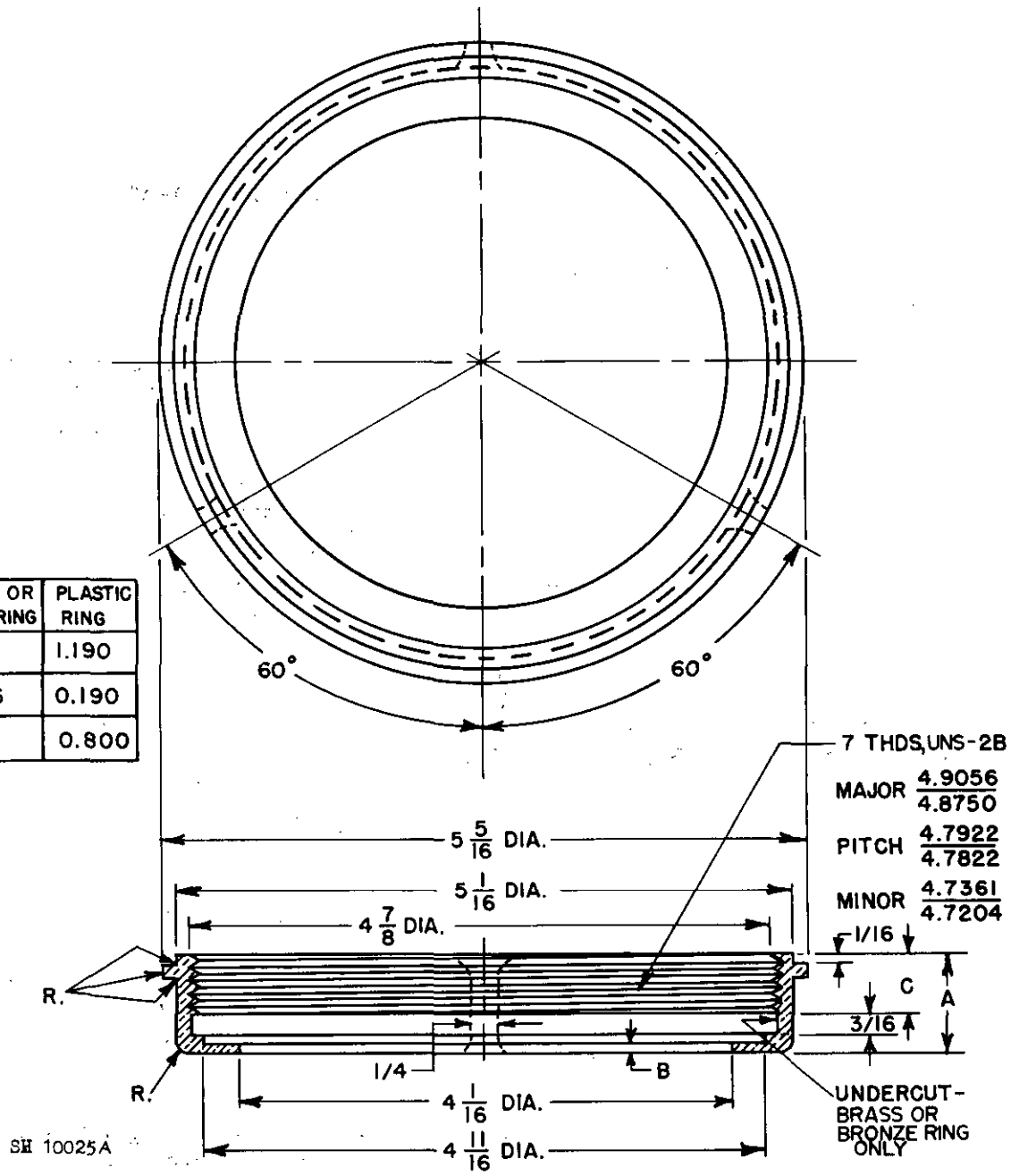


FIGURE 4. Securing ring.

Source: https://assist.dla.mil -- Downloaded: 202610-18T08:45Z
Check the source to verify that this is the current version before use.

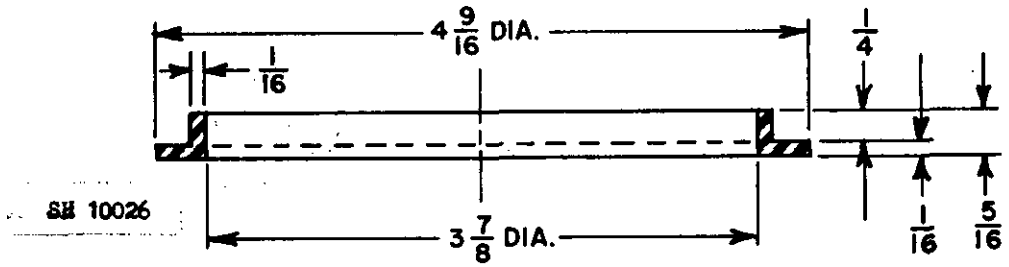


FIGURE 5. Globe centering gasket.

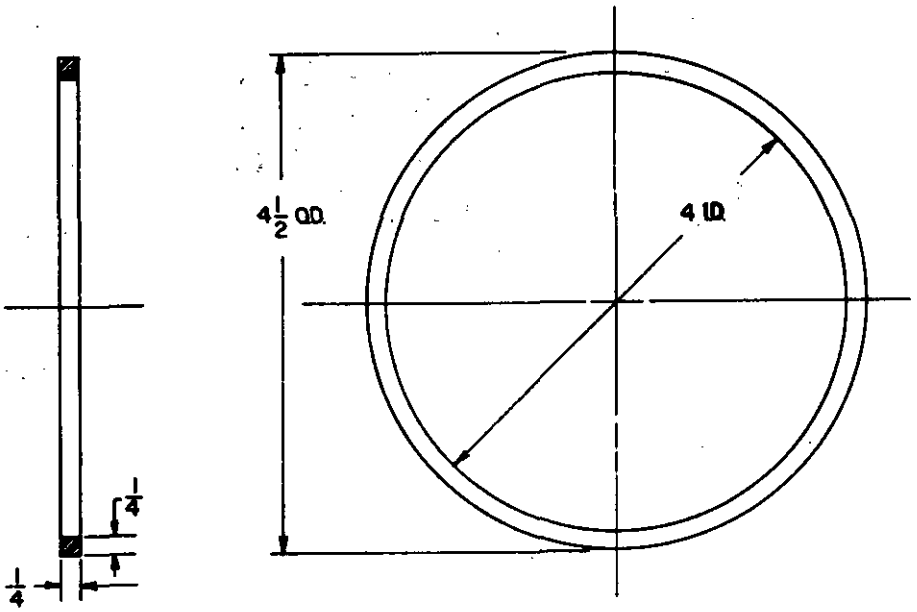


FIGURE 6. Enclosure gasket.

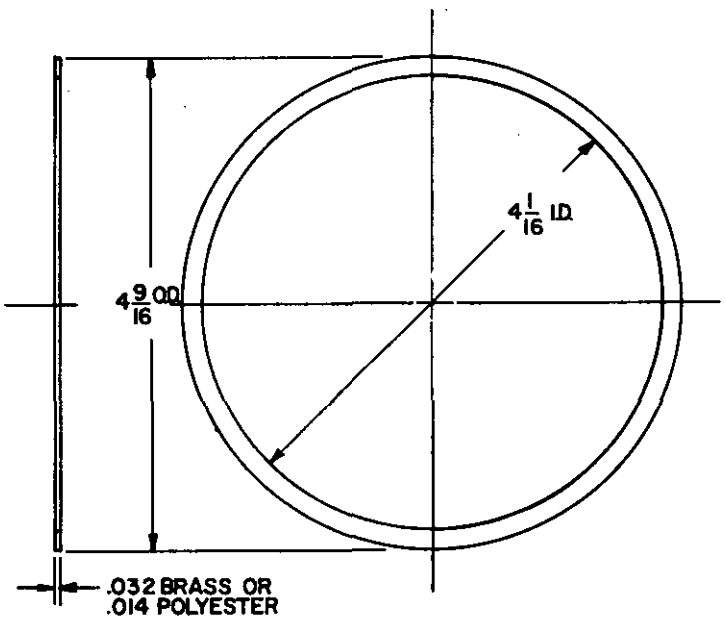


FIGURE 7. Slip ring.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:45Z
Check the source to verify that this is the current version before use.

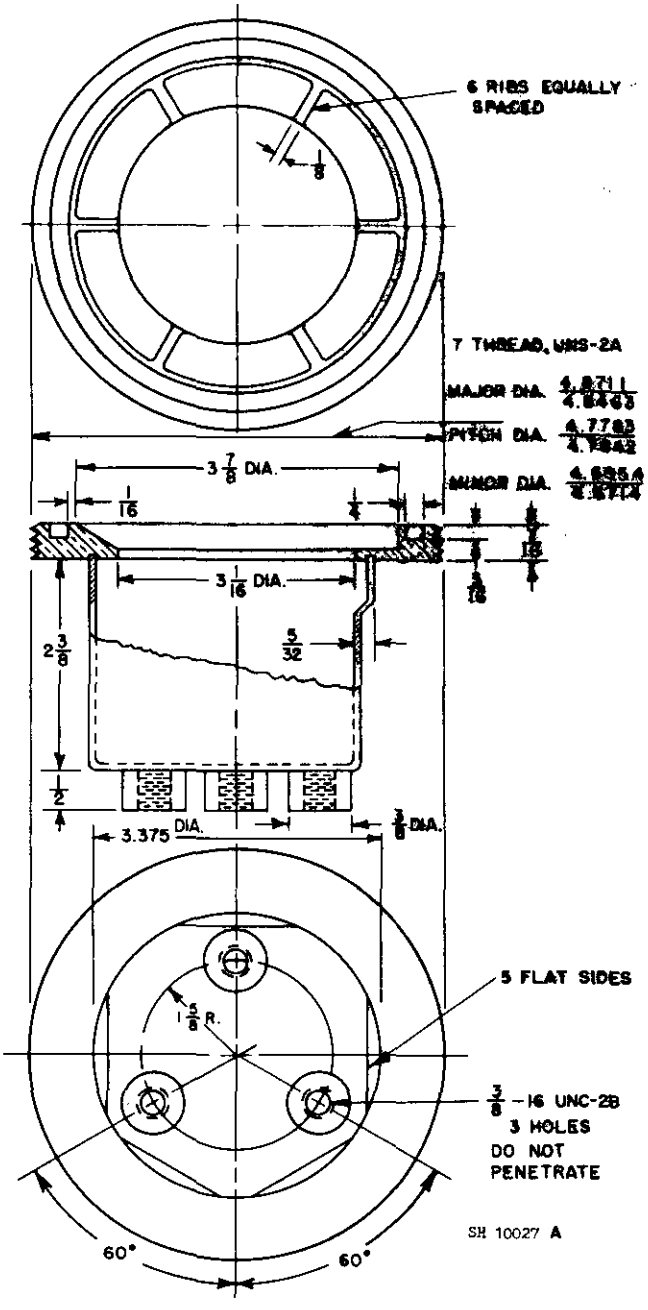
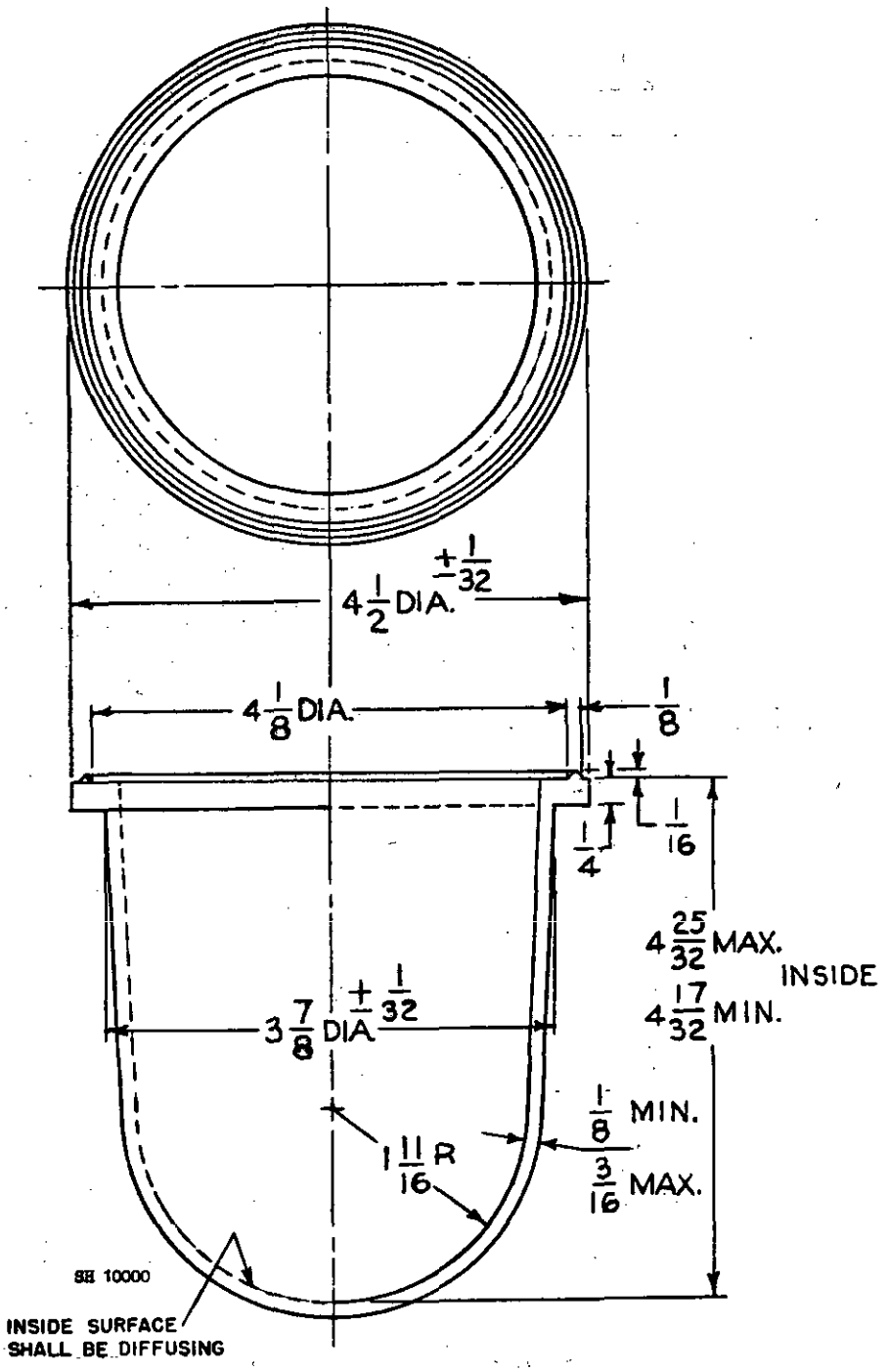


FIGURE 8. Enclosure assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:45Z
Check the source to verify that this is the current version before use.



Material - Tempered glass

Color:

Symbol 92.2 - Clear transparent/diffusing.
Symbol 93.2 - Red transparent/diffusing.
Symbol 93.4 - Blue transparent/diffusing
Symbol 93.5 - Yellow transparent/diffusing

Transmittance (symbol 92.2):

Clear transparent/diffusing - 80 percent minimum.

FIGURE 9. Globe.

Source: https://assist.dla.mil -- Downloaded: 202610-18T08:45Z
Check the source to verify that this is the current version before use.

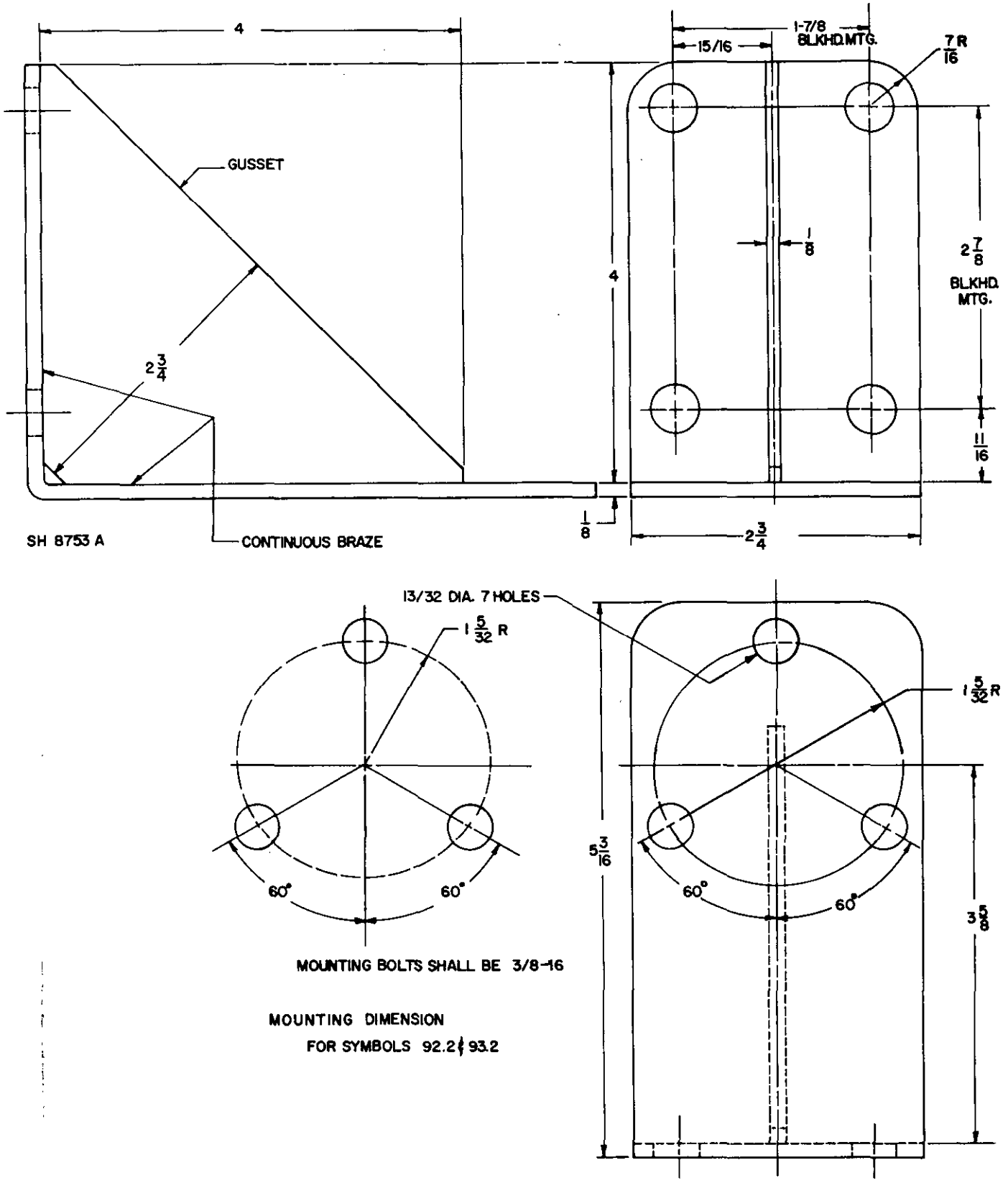
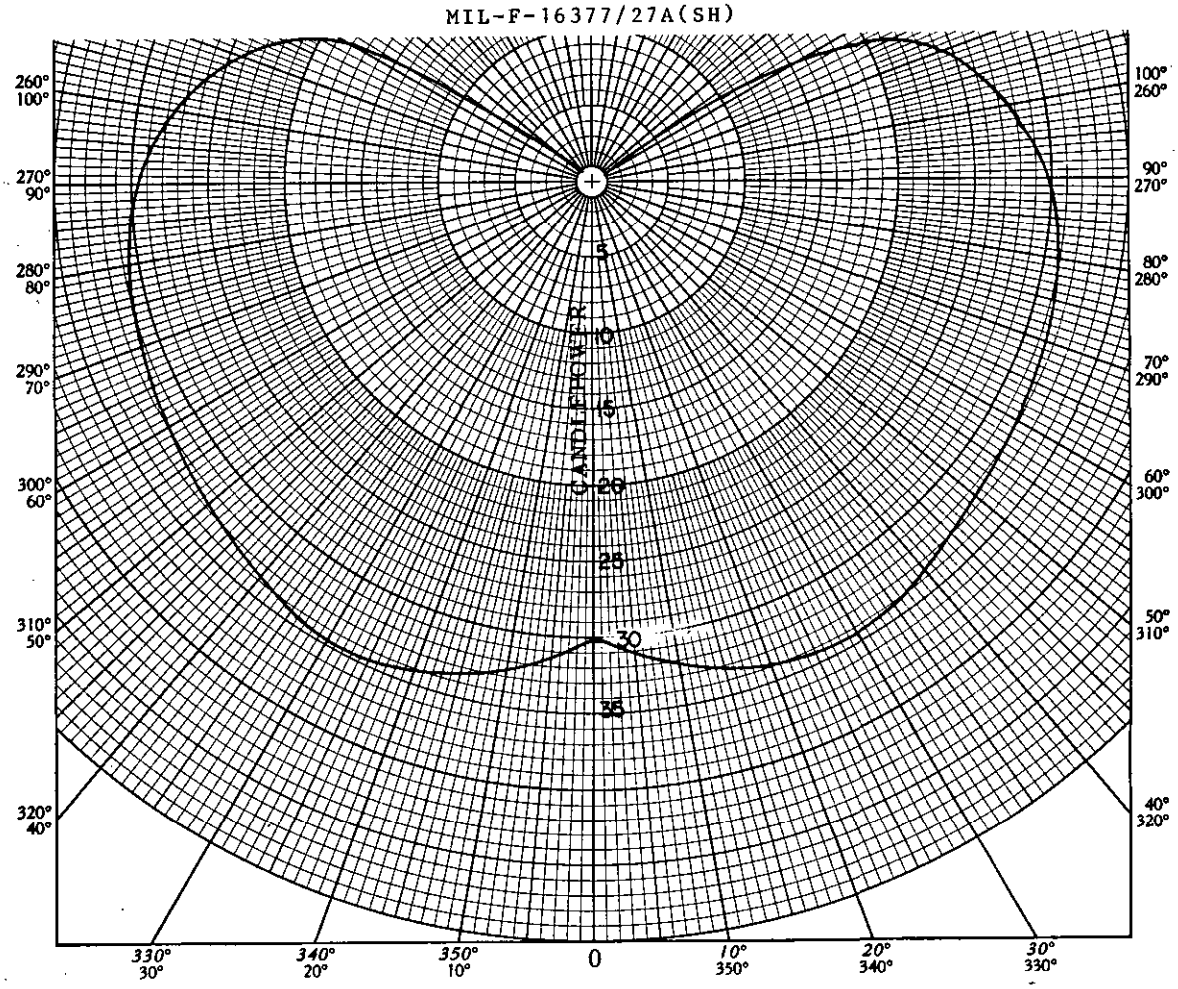


FIGURE 10. Mounting dimensions and bracket for bulkhead mounting. (Metal bracket only)

Source: https://assist.dla.mil -- Downloaded: 202640-18T08:45Z
Check the source to verify that this is the currentversion before use.



ANGLE	CANDLEPOWER SYMBOL 92.2
0	30
5	32
10	32.5
15	33
20	33.5
25	34
30	34
35	34
40	33
45	32
50	32
55	32
60	31.5

ANGLE	CANDLEPOWER SYMBOL 92.2
65	31
70	31
75	31
80	30.5
85	30
90	30
95	28
100	27.5
105	27
110	25
115	23
120	12.5
125	2

Photometric test data:

Distance: 10 feet.
Voltage: 120 volts.
Ambient temperature: 76°F.
Lamp: 50 watts, rough service conforming to MS15586-2, furnished by laboratory for test, shall be used.

FIGURE 11. Candlepower distribution curve.

Source: https://assist.dla.mil -- Downloaded: 202610-18T08:45Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Type: II

Class: 2

Symbol 92.2: Fixture for use with one 50-watt white incandescent lamp and white (clear transparent) diffusing globe.

Symbol 93.2: Fixture for use with one 50-watt white incandescent lamp and red (transparent diffusing) globe.

Symbol 93.4: Fixture for use with one 50-watt white incandescent lamp and blue (transparent diffusing) globe.

Symbol 93.5: Fixture for use with one 50-watt white incandescent lamp and yellow (transparent diffusing) globe.

Design:

Electrical rating: 120 volts, 50-watts.

Dimensions and configuration: See figures 2 through 9.

Enclosure assembly (includes box, threaded flange, tapped pads, and support plate): See figure 8. Material shall be brass, or a gray plastic, type MAI-60 in accordance with MIL-M-14.

Insulator for brass box assembly only: Plastic.

Gasket: Rubber, 1/4 inch square cross section by 4 inch I.D. by 4-1/2 inch O.D.

Lampholder: MIL-L-970/6, two contact.

Shockmounts: Brass, MIL-F-16377/41-012, or MIL-F-16377/41-015.

Centering gasket: Molded rubber, see figure 5.

Globes:

Symbol 92.2: White (clear transparent) diffusing. See figure 9.

Symbol 93.2: Red transparent diffusing. See figure 9.

Symbol 93.4: Blue transparent diffusing see figure 9.

Symbol 93.5: Yellow transparent diffusing see figure 9.

Slip ring: Brass, 4-9/16 inch o.d, by 4-1/16 inch o.d. by 1/32 inch thick.

Securing ring: Cast brass, bronze, or gray plastic type MAI-60 in accordance with MIL-M-14.

Bulkhead mounting bracket: When specified in the contract, a brass, corrosion resistant steel, or gray plastic type MAI-60 mounting bracket in accordance with MIL-M-14. The metal brackets shall be as shown on figure 3. Plastic brackets shall have mounting dimensions as shown on figure 3. Thickness of material shall be as necessary to meet the requirements of specified herein. Metal inserts are required on the mounting holes of plastic brackets.

Lamp: Fixtures shall be designed to accommodate a 50-watt white lamp conforming to MS15586-2. Lamp is not furnished.

Distribution curves Symbol 92.2: Candlepower (see figure 11).

Weight (Symbols 92.2 and 93.2): 2 pounds, 13 ounces max.

Labels: Labels shall be furnished in accordance with MIL-F-16377. One type IV label shall be applied as shown on figure 2. The information to be shown on the type IV label shall be:

"INSTALL 50-WATT LAMP ONLY.
LAMP: NSN6240-00-143-3070
WHITE GLOBE: NSN6210-00-702-5773
RED GLOBE: NSN6210-00-914-4152

Identification plate: Brass, stamped or engraved.

QUALITY ASSURANCE: Quality assurance shall be as specified in MIL-F-16377 and as follows:

First article inspection:	
Complete fixtures:	See table I.
Globes:	See table II.
Quality conformance:	
Comparison inspection:	Same as the first article inspection.
Inspection of product for delivery:	
Examination:	See MIL-F-16377.
Examination for preparation of delivery:	See MIL-F-16377.
Quality conformance tests:	See MIL-F-16377.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:45Z
Check the source to verify that this is the current version before use.

TABLE I. First article and comparison inspection for complete fixtures.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.4.1.1	4.8.4	
Mechanical shock	3.7.5.6	4.8.17.3	8 ft-lbs
Thermal shock	3.7.5.7	4.8.17.4	
Breakage pattern	3.7.5.9	4.8.17.6	
Color and luminous transmittance	3.7.3.1	4.8.6.3	Symbol 93.2 only
Light output	3.7.1	4.8.6.1	Symbol 92.2 only Candlepower
Magnetic permeability	3.4.8	4.8.18	

TABLE II. First article and comparison inspection for globes.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.5.1	4.8.4	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Light output	3.7.1	4.8.6.1	Symbol 92.2 only Candlepower
Mechanical shock	3.7.5.6	4.8.17.3	8 ft-lbs
Thermal shock	3.7.5.7	4.8.17.4	
Breakage pattern	3.7.5.9	4.8.17.6	
Color and luminous transmittance	3.7.3.1	4.8.6.3	Symbol 93.2 only

General Information: (a) Assignment of Military part numbers and general information is shown in table III.
(b) Fixtures symbols 92.2 and 93.2 may be adapted for bulkhead mounting by using the bracket shown on figure 3.

TABLE III. Part number and general information

Military part number M16377/27-	Item name	National stock No.(NSN)	Supersedes part number
92.2	Complete fixture, Symbol 92.2 Consisting of: Body	6210-00-299-7703	9000-S6401-74218
	and White (clear transparent diffusing) globe	6210-00-702-5773	Symbol 92.2
93.2	Complete fixture, Symbol 93.2 Consisting of: body	6210-00-299-7703	9000-S6401-74218
	and red transparent diffusing globe	6210-00-914-4152	Symbol 93.2
001	White (clear transparent diffusing) globe	6210-00-702-5773	MS17802-20
002	Red (transparent diffusing) globe	6210-00-914-4152	MS17802-21
003	Mounting bracket	5340-00-391-1073	
004	Body (complete less globe and mounting bracket)	6210-00-299-7703	
005	Securing ring	6210-00-399-6207	
006	Blue(transparent/diffusing)globe		
007	Yellow (transparent/ diffusing globe		

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6210-N575)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:45Z
Check the source to verify that this is the current version before use.